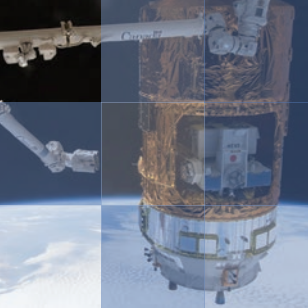




Inverse Trigonometric Functions

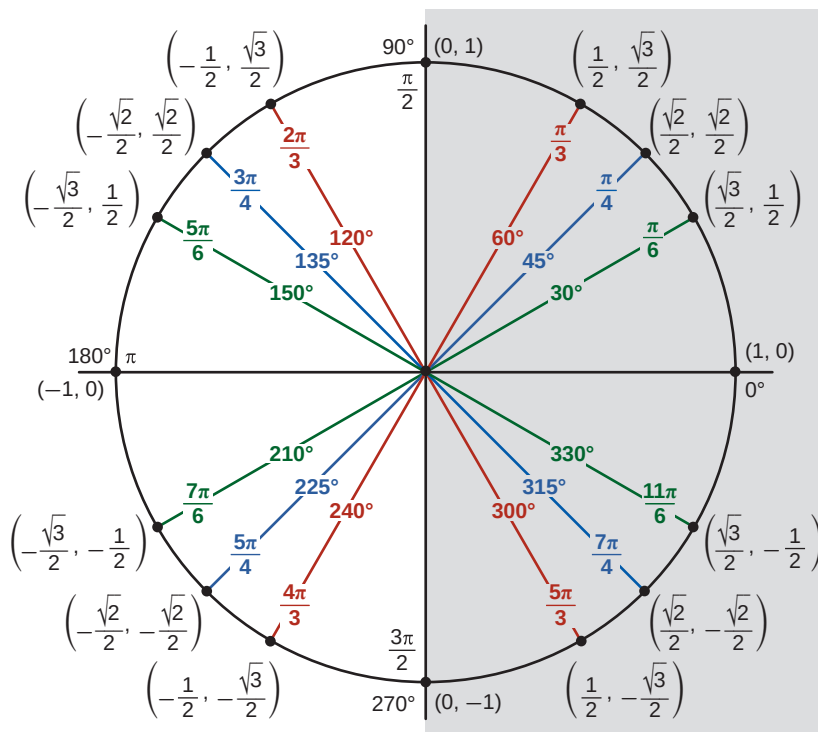
A trigonometric function has an inverse when its domain is restricted so that it is one-to-one.

Defining the Inverse Function of $y = \sin x$

Definition

In a **one-to-one function**, every element of the domain corresponds to exactly one element of the range.

When you look at the unit circle, you can see several values of $\sin x$ for which there is more than one input that results in that same output. Look at $\sin \frac{\pi}{6} = \frac{1}{2}$ and $\sin \frac{5\pi}{6} = \frac{1}{2}$ as an example. Because the inputs $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ both give the output $\frac{1}{2}$, $y = \sin x$ is not a one-to-one function. However, when the inputs are restricted to values from $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, $\sin x$ is strictly increasing.



In this interval, the values of $\sin x$ do not repeat. The function is therefore one-to-one when its domain is restricted to values between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, inclusive. So it is possible to define an inverse function of $\sin x$ on this interval.

Definition

The **arcsine (arcsin) function** $y = \arcsin x$ if and only if

$$\sin y = x$$

where $-1 \leq x \leq 1$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Another way of stating this definition is “The inverse function of $y = \sin x$ is $y = \arcsin x$ as long as $\sin x$ is restricted to input values from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.”

► **Think About It** If two functions are inverses of each other, then their inputs and outputs are switched. The input of the sine function is an angle measure, and the output is a real number between -1 and 1 . The input of the arcsine function is a real number between -1 and 1 , and the output is an angle measure.

EXAMPLE 1

Determine the exact value of $\arcsin\left(\frac{-\sqrt{3}}{2}\right)$ in radians.

SOLUTION

You want to determine an angle θ such that $\sin \theta = \left(\frac{-\sqrt{3}}{2}\right)$.

Look on the unit circle in the interval from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ and locate the angle with a sine value of $\frac{-\sqrt{3}}{2}$.

The angle $-\frac{\pi}{3}$ has a sine value of $\frac{-\sqrt{3}}{2}$, so $\arcsin\left(\frac{-\sqrt{3}}{2}\right) = -\frac{\pi}{3}$. ■

Defining the Inverse Function of $y = \cos x$

The cosine function can also be restricted so that it is one-to-one. From 0 to $\frac{\pi}{2}$, the interval indicated by the shaded region, cosine is a decreasing function as it takes on all values from -1 to 1 . It is therefore one-to-one from 0 to $\frac{\pi}{2}$. So cosine has an inverse function when its domain is restricted to this interval.

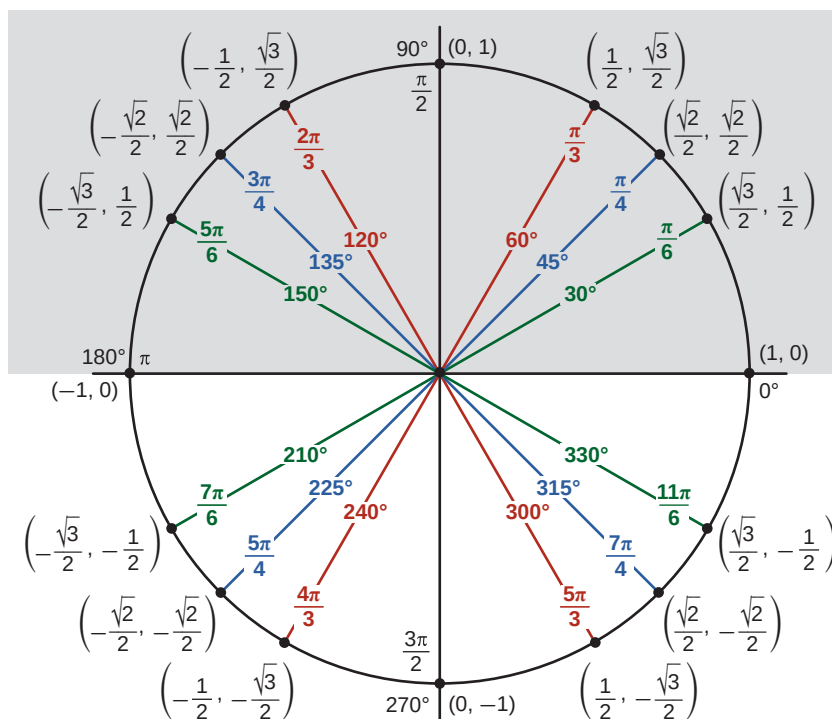
Definition

The **arccosine (arccos) function** $y = \arccos x$ if and only if

$$\cos y = x$$

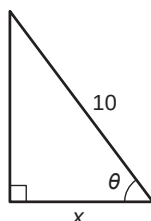
where $-1 \leq x \leq 1$ and $0 \leq y \leq \pi$.

You can set up equations involving inverse trigonometric functions based on triangle side lengths and angles.



EXAMPLE 2

Write θ as a function of x .



SOLUTION

Side length x is adjacent to angle θ , and the hypotenuse of the triangle is 10. So start with $\cos \theta$, which involves the ratio of the adjacent side length to the length of the hypotenuse. Then use the definition of arccosine.

$$\cos \theta = \frac{x}{10}$$
$$\theta = \arccos\left(\frac{x}{10}\right)$$

You can use reference angles to evaluate expressions involving inverse trigonometric functions. ■

► **Think About It** When you are asked to write one variable as a function of another variable, the function equation should end up with the first variable isolated on one side. In this case, θ is the variable to be isolated.

EXAMPLE 3

Use a calculator to determine $\arcsin(0.6)$ to the nearest hundredth of a radian.

SOLUTION

Make sure that your calculator is in radian mode. Then enter the expression **arcsin(0.6)** and press **ENTER**.

$$\arcsin(0.6) \approx 0.6435 \quad \blacksquare$$

► **Think About It** On a scientific calculator, enter the value, then select the **Inv** or **Shift** button and then the trig function. For instance, **0.6**, then **Inv**, and then **sin**⁻¹.